

Indian Institute of Technology Kharagpur

Department of Mathematics

Semester-8(DC)

Big-data Analysis

(LTP- 3-0-0,CRD- 3, 40 hrs)

Prerequisites :

Probability & Statistics (MA20205) and Linear Algebra

Syllabus :

1. Recap: Linear Algebra - 3 Lectures
2. Tools from Multivariate Analysis - 3 Lectures
3. Statistical Learning - 3 Lectures
4. Classification - Disc. Analysis, Logistic Regression, Multinomial Logistic Regression - 4 Lectures
5. Multiple-Linear model and Regularization (PCR,PLS, RIDGE)-5 Lectures
6. High dimensional Regression (LASSO, ELASTIC NET)-5 Lectures
7. Clustering: Mixture Model , K-Means, DBSCAN-5 Lectures
8. Stream data analysis: CUSUM, Change point - 3 Lectures
9. Text Analytics - Latent Dirichlet Allocation (Topic Model)- 3 Lectures
10. Random Graph-Data Analysis: 4-Lectures
11. Image data Analysis: 2-Lectures

Reference:

1. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning* (Vol. 112, p. 18). New York: springer.
2. Strang, G. (2019). *Linear algebra and learning from data*. Cambridge: Wellesley-Cambridge Press.
3. Blum, A., Hopcroft, J., & Kannan, R. (2020). *Foundations of data science*. Cambridge University Press.

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